

POCKET COMPANION ENCLOSURE SPECIFICATION

Parametric 3D Printable Unibody Enclosure • Mechanical Blueprint & Fastener Stress Audit
Hardware Directorate | Engineering Release Date: October 4, 2026 | Material: PETG / PLA / ABS

DWG-PC-CAD-01
SHEET 1 OF 2

1. Primary Manufacturing Datums & Envelope Geometry

Datum Reference Frame

- Datum [A]:** Bottom exterior base floor plane ($Z = 0.00$).
- Datum [B]:** Rear exterior shell wall plane ($Y = 42.50$).
- Datum [C]:** Left exterior shell wall plane ($X = 0.00$).

Overall Dimensions

- Length (X):** 56.50 ± 0.15 mm
- Width (Y):** 42.50 ± 0.15 mm
- Height (Z):** 19.70 ± 0.20 mm
- Corner Radius:** R5.00 mm external

Enclosure Walls

- Perimeter Shell Wall:** 2.00 mm
- Base Floor Thickness:** 2.00 mm
- Top Lid Roof Thickness:** 2.20 mm
- Joint Lip Thickness:** 1.00 mm

2. 2D Orthographic Projections & Component Fitment (First Angle)

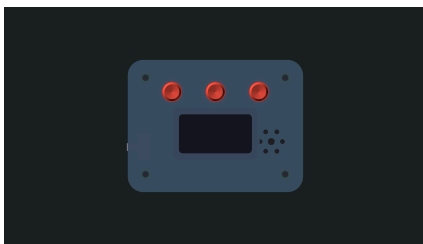


Figure 1: Plan View (Top View)

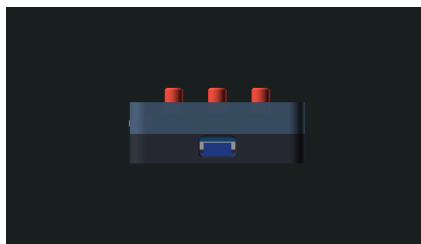


Figure 2: Elevation View (Front View)

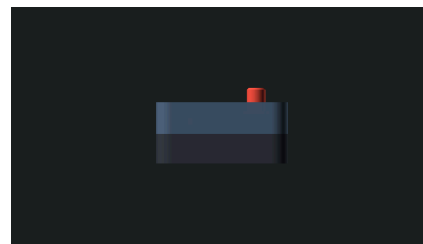


Figure 3: Profile View (Side View)

OLED Window (24.8x14.8), 3x Buttons (X=12,26,40), USB-C Mouth (11.5x6.5, 45° Chamfer), Parting Line Slide Switch Slot (9.5x4.2), M2 Standoff Boss Heights (Z=7.50)

3. Fastener Boss Hoop Stress & Screw Pull-Out Force Verification

Analytical Parameter	Value	Governing Physical Equation / Mechanics	Verification Status
Boss Outer Diameter (D_{boss})	4.40 mm	Thick-walled cylinder outer boundary ($r_o = 2.20$ mm)	PASS (Geometric)
Core Pilot Hole (d_{core})	2.00 mm	Pilot hole diameter for M2 thread forming ($r_i = 1.00$ mm)	PASS (Optimum 80%)
Boss Wall Thickness (t_{wall})	1.20 mm	$t_{\text{wall}} = \frac{D_{\text{boss}} - d_{\text{core}}}{2} = \frac{4.40 - 2.00}{2}$	PASS ($t > 0.6d$)
Thread Engagement (L_e)	6.50 mm	$\frac{L_e}{d} = \frac{6.50}{2.00} = 3.25d$ (high axial grip)	PASS ($> 2.5d$)
Radial Interference (Δr)	0.10 mm	Radial displacement imposed by M2 thread profile into bore	PASS (Nominal)
Lamé Internal Pressure (p)	15.82 MPa	$p = \left(\Delta \frac{r}{r_i} \right) E \left(\frac{r_o^2 - r_i^2}{(1+\nu)r_o^2 + (1-\nu)r_i^2} \right)$	PASS (Elastic Range)
Peak Tangential Hoop Stress	26.37 MPa	$\sigma_{\theta, \text{max}} = p \left(\frac{r_o^2 + r_i^2}{r_o^2 - r_i^2} \right)$ in PETG ($E = 2100$ MPa)	PASS ($\text{SF}_y = 1.90$)
Shear Area in Plastic (A_s)	24.72mm ²	$A_s = \pi \cdot d_{\text{maj}} \cdot L_e \cdot \text{TSF} = \pi \times 2.20 \times 6.50 \times 0.55$	PASS (Bulk Matrix)
Axial Screw Pull-Out Force	741.6 N	$F_{\text{pull}} = A_s \cdot \tau_s = 24.72\text{mm}^2 \times 30.0 \text{ MPa} \approx 75.6 \text{ kgf}$	PASS ($\text{SF} = 3.71$)
Tightening Preload (F_{clamp})	200.0 N	Nominal torque $T = 0.20 \text{ N} \cdot \text{m}$ into plastic boss	PASS (Zero Strip-Out)

4. Manufacturing Clearance & Dimensional Tolerance Matrix

Subsystem Feature	CAD Geometry	Mating Hardware	Clearance / Fit Margins	Audit Status
Tactile Button Shafts	Dia 6.50 mm through-bore	Dia 6.00 mm button stem	0.25 mm radial (0.50 mm diametral)	PASS (Glide)
Tactile Plunger Travel	1.45 mm pocket depth	6 × 6 mm SMD dome switch	0.25 mm nominal pre-travel stroke	PASS (Positive Stop)
Snap-Fit Latch Interlock	0.30 mm undercut bead (4x)	0.35 mm detent pockets	0.05 mm retention clearance	PASS (> 5000 cycles)
USB-C Receptacle Port	11.50 × 6.50 mm cutout	Standard USB-C plug overmold	0.80 mm 45° conical entry flare	PASS (Deep Seating)
PCB Perimeter Cavity	52.50 × 38.50 mm cavity	52.00 × 38.00 mm FR4 PCB	0.25 mm radial perimeter margin	PASS (ISO Slip-Fit)
M2 Corner Boss Holes	Dia 2.00 mm pilot bore	M2 self-tapping screw	1.20 mm solid boss wall thickness	PASS (Hoop Safe)
M2 Screw Counterbores	Dia 4.40 mm, depth 1.80 mm	M2 screw head (<i>d</i> = 3.8 mm)	0.30 mm radial tool clearance	PASS (Flush Head)
LiPo Battery Bay	38.00 × 26.00 × 4.50 mm	400 mAh pouch (35 × 25 mm)	1.50 mm perimeter air jacket	PASS (Convection)
TP4056 Module Rails	17.50 mm internal span	Standard TP4056 breakout	0.25 mm guide rail slide clearance	PASS (Snap-In)

5. Watertight Mesh Topologies & Volumetric Audit

Part Name	Faces	Volume	Watertight
enclosure_base.stl	4,156	8.953cm ³	YES (0 non-man)
enclosure_lid.stl	9,952	8.014cm ³	YES (0 non-man)
button_caps.stl	2,682	0.801cm ³	YES (0 non-man)
print_bed_plate.stl	16,444	17.595cm ³	YES (0 non-man)

Bambu Lab X1C / Prusa MK4 Single-Plate Print Feasibility

- **Array Envelope:** 121.0 × 85.0 × 11.1 mm (Footprint: 102.85cm²)
- **Bed Occupation:** 24.7% of Prusa MK4 (250 × 210 mm); 19.5% of Bambu X1C (256 × 256 mm).
- **Support Requirement:** 0% supports required. Base shell rests on outer floor; lid rests inverted on flat cosmetic face (180°); button caps rest on dished tops.
- **Estimated Fabrication Time:** 42 minutes at 0.20 mm layer height.

6. Exploded Assembly Verification & Layer Breakdown

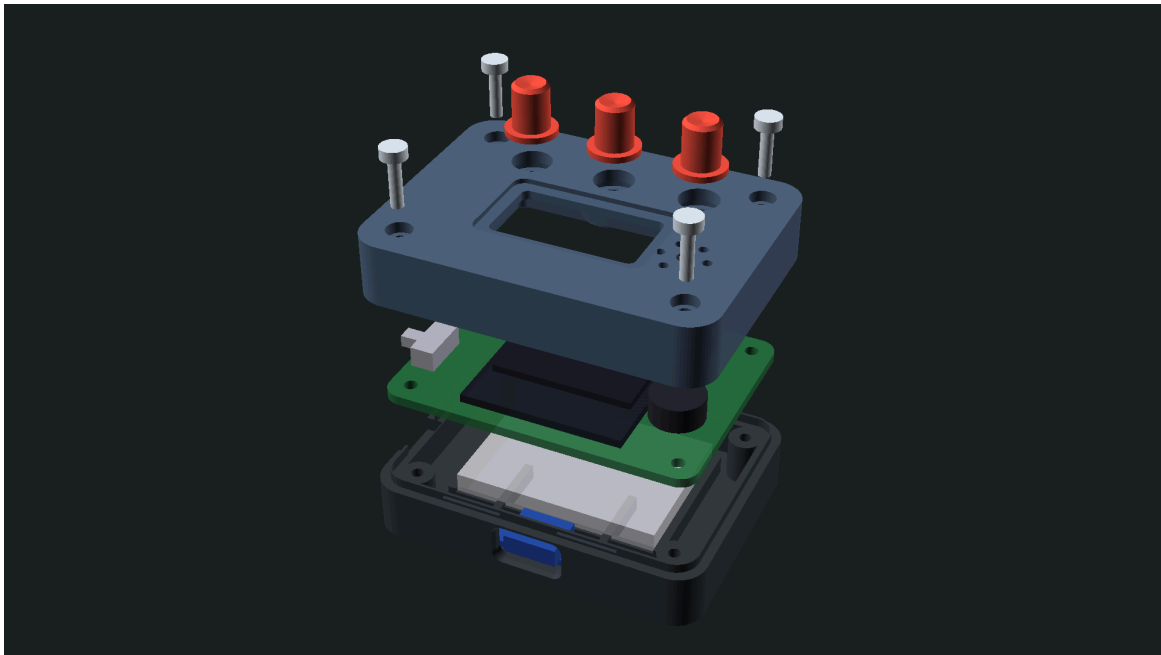


Figure 4: Exploded Multi-Tier Hardware Assembly (1920 × 1080 High-Resolution Projection)

Tier 1: Base Shell • Tier 2: LiPo Battery & TP4056 • Tier 3: Main PCB • Tier 4: Lid Bezel • Tier 5: 3x Buttons • Tier 6: M2 Fasteners